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Louis Fostier

Education

- 02/2026–
Present **Postdoctoral Researcher, Modeling of adipose tissue morphogenesis, ENERGENCE project**, team *MUSCLEES*, INRIA, Paris, France.
Keywords: Computational biology ; Individual-based model ; Sensibility analysis ; Metamodeling ; Tissue morphogenesis
- 11/2022–
10/2025 **PhD student on *Analysis and calibration of structured population dynamics models: application to cell populations***, team *BIOS*, INRAE, PRC, Nouzilly, France.
Mathematical modelling in biology, Well-posedness and long-time analysis of size-structured population dynamics models, numerical methods for inverse problem.
Keywords: Size-Structured Population Dynamics ; Asymptotic Behavior ; Bifurcation Analysis ; Inverse Problem ; PINNs ; Reproductive Biology
- Advisors: Frédérique Clément (team MUSCA, INRIA), Romain Yvinec (team BIOS, INRAE) and Violette Thermes (LPGP, INRAE).
- 2020–2022 **Master of Applied Mathematics**, *Université de Rennes 1*.
Scientific computing, modelling, numerical analysis. Obtained with very good honors.
- 2017–2020 **Bachelor of Mathematics**, *Université de Rennes 1*.
Obtained with very good honors.

Internship and Hackaton

- 2023 **6 weeks hackaton on *Scientific machine learning for equation discovery***, *Centre d'Été Mathématique de Recherche Avancée en Calcul Scientifique*, Marseille, France.
- 2022 **M2 internship on *Modelling of the female fish germ cell population***, team *MUSCA*, INRIA, Palaiseau, France.
- Advisors: Frédérique Clément (team MUSCA, INRIA) and Romain Yvinec (team BIOS, INRAE)
- 2021 **M1 internship on *Scientific computing and modelling for biofilms***, *IRMAR, CNRS*, Rennes, France.
- Advisors: Fabrice Mahé (IRMAR, CNRS) and Bénédicte Martin (NUMECAN, INSERM).

Publications

Journal Article

- 2026 **Fostier L., Dauger A., Yvinec R., Ribot M., Audebert C., Soula H.**, Rapid cell turnover to model adipocyte size distribution, in *Journal of Theoretical Biology*.
<https://www.sciencedirect.com/science/article/pii/S0022519325002954>
- 2025 **Clement F., Fostier L., Yvinec R.**, Bifurcation analysis of a size-structured population model: Application to oocyte dynamics and ovarian cycle, in *SIAM Journal on Applied Dynamical Systems (SIADS)* (Selected by scientific committee for ReproHighlights GdR Repro 2025) .
<https://doi.org/10.1137/24M1705147>
- 2022 **Chathoth K., Fostier L., Martin B., Baysse C., Mahé F.**, A Multi-Skilled Mathematical Model of Bacterial Attachment in Initiation of Biofilms, in *microorganisms*.
<https://www.mdpi.com/2076-2607/10/4/686>

Proceeding

- 2025 **Fiorini C., Flint C., Fostier L., Franck E., Hashemi R., Michel-Dansac V., Tenachi W.**, Generalizing the SINDy approach with nested neural networks, in *ESAIM: Proceedings and surveys*.
<https://www.esaim-proc.org/articles/proc/abs/2025/04/proc25081009/proc25081009.html>

Preprint

- 2026 **Fostier L., Lesage M., Thermes V., Clément F., Yvinec R.**, Data-driven PINN inference of oocyte dynamics in fish ovaries.
<https://inria.hal.science/hal-05536280>

Communications in conference and seminar

Oral communications

- 2025 Math seminar of the Laboratoire de Mathématiques Appliquées du Havre, November 27, Le Havre
2025 Math Bio seminar of Institut Denis Poisson, September 24, Tours
2025 Differential Equations and Applications to Biology, June 16-20, Le Havre
2025 Biennale de la SMAI, June 02-05, Carcans-Maubuisson
2025 Métaprogramme DIGIT-BIO seminar, December 12-13, Lyon
2024 Métaprogramme DIGIT-BIO seminar, December 12-13, Lyon
2024 Congrès des Jeunes Chercheur.e.s en Mathématiques Appliquées, October 28-30, Lyon
2024 Biology and Medicine Through Mathematics Conference, May 15-17, Richmond USA
2024 Math bio seminar, March 22, Duke University USA
2023 Journées Math-Bio-Santé, November 27 - December 1, Paris Marne La Vallée
2023 Math Bio seminar of Institut Denis Poisson, July 13, Orléans
2023-2025 PhD student seminar of the research unit Physiologie de la Reproduction et des Comportements
2023-2024 PhD student seminar week of Institut Denis Poisson

Posters

- 2025 Journées Math-Bio-Santé, November 05-07, Montpellier
2024 European Conference on Mathematical and Theoretical Biology, July 22-26, Toledo
2024 Journées Math-Bio-Santé, June 24-28, Nantes
2023 Journées reprodsciences, 3-5 May, Paris

Teaching and tutoring

- 2024 **L1 classes of Calculus and mathematical reasoning**, Université de Tours, France.
30 hours
2022 **L1 classes of Algebra and Analysis**, Université de Tours, France.
54 hours
2021 **Assistance and pedagogical support for disabled student**, Université de Rennes 1, France.
2020 **Educational tutoring for students in L3 mathematics**, Université de Rennes 1, France.

Popularisation

- 2024 **IA workshop**, *Seminar of the INRAE research unit Physiologie de la Reproduction et des Comportements*.

Research stays abroad

- 2024 **Duke University**, *March 11 – May 10, 2024*, INRIA funding (€10k) – Sabbatical program: Dissemination of PhD research work, support from Kevin Flores (NC State University) to explore BINN (Biologically Informed Neural Network) methods.

Awards

- 2024 Best poster award - European Conference on Mathematical and Theoretical Biology , Toledo

Computer skills

Basic C++

Intermediate C, FORTRAN, Matlab, FreeFem++

Advanced PYTHON

Referees

Dr. Frédérique Clément

Research Director, head of MUSCA team

Inria Saclay

✉ frederique.clement@inria.fr

Dr. Magali Ribot

University professor

Institut Denis Poisson

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Dr. Romain Yvinec

INRAE Research Director, head of BIOS team

INRAE Val-de-Loire

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